

UHI9: Impermanent Loss & Yield Systems

Hookathon Research & Idea Guide

Uniswap Hook Incubator | Cohort 9 | Hookathon: May 25, 2026 | Demo Day: June 19, 2026

Theme Overview

UHI9 focuses on designing mechanisms that protect LPs from impermanent loss while unlocking sustainable onchain yield. Impermanent loss remains one of the biggest barriers to LP participation. If hooks can turn Uniswap into a yield-protected AMM layer, liquidity can become stickier, LPs can earn more predictable returns, and Uniswap can compete more effectively on volatile pairs.

Key Problem Areas

- LPs lose value when asset prices diverge — this is impermanent loss (IL)
- Out-of-range liquidity sits idle, earning nothing
- Fee revenue is unpredictable and often insufficient to offset IL
- LPs have no native tools for hedging or yield enhancement inside Uniswap

Official Prompt Examples from Uniswap

- Impermanent-loss insurance hooks that offset losses via dynamic rebalancing or premium accrual
- YieldBasis-style fixed-income systems that create predictable returns for LPs
- Delta-neutral hooks that hedge LP exposure using per-block swaps, options, or external hedging protocols
- Fee-smoothing hooks that automatically reinvest or distribute yield over time
- Cross-pool hedging routers that match opposing liquidity positions across correlated pairs

15 Examples from Past Cohorts

These 15 projects from UHI1–UHI8 are the strongest precedents for the UHI9 theme. Share these with students as inspiration for scope, mechanism, and execution quality.

IL Protection & Hedging

Project	Cohort	What It Does
FlexFee	UHI1	Protects LPs from IL using dynamic fees based on volatility, calculated via Brevis ZK proofs. One of the earliest and cleanest implementations of volatility-linked fee adjustment.
LVR & IL Hedge Hook	UHI2	Attempts to bring DeFi to a risk-neutral state by combining dynamic fees with delta-gamma neutrality. Explicitly targets both LVR and IL in a single hook.
Gainswap: Impermanent Gain	UHI3	Enables liquidity borrowing to short-sell impermanent loss — effectively transforming IL into impermanent gain. Creative and novel mechanism.
DynamicShield Hook	UHI3	Dynamically adjusts fees, converts inactive assets into safe tokens during downturns, and reinvests during market recoveries to protect LPs from IL.
xstreamly hook	UHI4	AI-powered smart hedging for LPs using volatility predictions and perpetual futures to minimize IL and maximize risk-adjusted returns.
Protected Pool Hook	UHI4	Protects LPs from IL in depeg scenarios by integrating Cork Protocol's Peg Stability Module as the hedge mechanism.
EquiGuard	UHI5	Mitigates traditional v3 IL through a reserve hedge, explicitly benchmarking against v2 IL exposure while retaining v3 fee yield.
Non Toxic Pools	UHI8	Claims to eliminate IL entirely through reverse-engineered Uniswap math to create a dynamic fee pool that is optimized for market makers.
Confidential IL Insurance Hook	UHI6	Confidential IL insurance using Fhenix FHE and EigenLayer AVS for private risk assessment and decentralized payout verification.
PILI	UHI7	Automated, privacy-preserving protection against impermanent loss for LPs, using encrypted position data.

Idle Liquidity & Yield on Out-of-Range Positions

Project	Cohort	What It Does
Idle Liquidity Yield Earning Hook	UHI2	The archetype for this category: deposits out-of-range LP liquidity into lending protocols (e.g. Aave/Morpho) to earn yield, then withdraws when price re-enters range.
Unicord	UHI2	Deploys all stablecoin pool liquidity to Morpho and executes on-demand withdrawals for trades. Maximizes capital efficiency for stable pairs.

Project	Cohort	What It Does
Idle Liquidity Optimizer Hook	UHI3	Auto-deploys out-of-range liquidity to lending protocols and returns it to the pool when prices move back into range, eliminating idle capital.
YieldSync Hook	UHI6	Auto-adjusts LST pool LP positions to compensate for yield drift in stETH/rETH positions, claiming 95%+ capital efficiency and 75% IL reduction.
FlexiPool	UHI6	When an LP position goes out-of-range, deposits into Aave to earn lending yield. When price returns to range, withdraws and reactivates for swaps.

25 Brainstormed Hook Ideas

The following 25 ideas are organized by approach. These range from beginner-accessible to advanced, and from closely aligned with the official prompt to more experimental. Students should use these as starting points — not finished specs.

Category 1: IL Protection & Insurance

1. Dynamic Rebalancing Hook — Automatically rebalances LP positions when price divergence hits a threshold, using protocol-owned reserves to cover the delta.
2. IL Insurance Premium Fund — LPs pay a small premium into a pooled mutual insurance fund. Payouts trigger when IL exceeds a user-set threshold at withdrawal. Unlike dynamic fees, this is an actual insurance pool.
3. Hedged Entry Hook — On LP deposit, automatically opens a small hedge via a perp protocol (GMX, Hyperliquid, etc.) to offset directional exposure.
4. Volatility-Gated LP Entry Hook — Uses an oracle to pause new LP deposits during high-volatility windows, preventing LPs from entering at statistically bad times.
5. Time-Weighted IL Smoothing Hook — Spreads IL realization across time rather than crystallizing it at withdrawal. Rewards longer-tenure LPs with lower effective IL.

Category 2: Fixed Income & Predictable Yield

6. YieldBasis-Style Tranche Hook — Splits LP positions into senior (fixed yield) and junior (variable) tranches. Senior LPs get guaranteed returns; junior absorbs IL.
7. Fixed APY Subscription Hook — LPs lock liquidity for a defined period and receive a guaranteed yield rate, funded by swap fees supplemented by external yield sources (Aave, Morpho).
8. Fee Accumulator + Vesting Hook — Fees don't distribute immediately but vest over time, rewarding long-term LPs and discouraging mercenary capital.

9. Yield Tokenization Hook — Splits LP positions into principal tokens and yield tokens (Pendle-style), allowing LPs to sell future yield upfront for immediate liquidity.
10. Boosted Yield Hook with veToken Integration — LPs who stake a governance token receive boosted fee share, creating stickier liquidity incentives.

Category 3: Delta-Neutral & Hedging Hooks

11. Per-Block Swap Hedging Hook — Executes micro-swaps each block to keep the LP's net delta near zero, neutralizing directional exposure continuously.
12. Options Collar Hook — On deposit, automatically buys a put and sells a call on-chain (via Lyra, Dopex, or Panoptic) to create a collar around the LP position.
13. Cross-Pool Hedging Router — Routes trades across correlated pairs (e.g. ETH/USDC + stETH/ETH) so opposing LP positions cancel out IL across pools.
14. Perp Funding Rate Arbitrage Hook — When funding rates on perp protocols are favorable, opens hedges that generate positive yield to offset IL simultaneously.
15. Gamma-Neutral Range Rebalancer — Dynamically adjusts LP range width based on realized vs. implied volatility to maintain a gamma-neutral stance.

Category 4: Fee Management & Yield Distribution

16. Auto-Compounding Hook — Collected fees are automatically reinvested into the LP position every block, compounding returns without gas overhead for LPs.
17. Fee Smoothing Hook — Converts lumpy fee distributions into continuous real-time streams to LP addresses, improving predictability.
18. Dynamic Fee Hook (Volatility-Linked) — Fee tier adjusts based on real-time volatility so LPs are always compensated proportionally for the risk they take.
19. MEV Capture + LP Redistribution Hook — Captures sandwich/arbitrage MEV and redistributes it to LPs as a yield supplement.
20. Liquidity Mining Replacement Hook — Replaces token emission incentives with protocol-fee-funded yield, making LP rewards sustainable without inflation.

Category 5: Experimental & Novel Approaches

21. JIT Defense Hook — Detects JIT liquidity attacks and penalizes or routes around them, protecting passive LPs from being front-run on every large swap.
22. Social Recovery LP Hook — LP positions have a guardian address that can rebalance or exit on behalf of the LP — useful for less active participants.
23. Cross-Chain IL Hedge Hook — Opens hedges on a cheaper chain (via LayerZero or Wormhole) while the LP position sits on mainnet.
24. LP Position NFT with Embedded Hedge — Each LP position is an NFT carrying its hedging parameters, making positions composable, transferable, and tradeable.

25. AI Oracle-Driven Range Hook — Uses an on-chain ML oracle to predict the optimal range width and rebalances proactively before IL accumulates.

Ideas That Have Not Been Attempted

After reviewing all ~300+ hooks across UHI1–UHI8, the following ideas from the brainstorm above appear to be genuinely unexplored. These represent the largest white spaces for UHI9 students.

1. Yield Tokenization Hook (Idea #9)

No past project has split LP positions into principal tokens and yield tokens natively inside a hook. Pendle does this at the protocol level for external yield-bearing tokens — but applying the same model to LP fee yield within a Uniswap v4 hook is entirely unexplored. This would allow LPs to monetize expected future yield immediately.

2. Fixed APY Subscription Hook (Idea #7)

The Fixed/Leverage Yield Hook (UHI3) attempted a tranche-based model, but no project has built a clean, lockup-based guaranteed APY product funded by fees plus external yield. This is one of the most commercially legible ideas in the theme — it mirrors how fixed-income products work in TradFi.

3. IL Insurance Premium Fund (Idea #2)

Most IL protection hooks adjust fees or rebalance positions. No project has built an actual pooled insurance fund where LPs pay premiums and can file claims when IL exceeds a threshold. This is structurally different — it separates the protection mechanism from the pool mechanics entirely.

4. Volatility-Gated LP Entry Hook (Idea #4)

Every past project reacts to volatility after it happens (by changing fees or rebalancing). No hook has gated new LP deposits based on predicted volatility — preventing LPs from entering during statistically bad windows in the first place. This is a proactive, rather than reactive, approach.

5. Time-Weighted IL Smoothing (Idea #5)

No project has attempted to spread IL realization across time rather than crystallizing it at withdrawal. This could dramatically change LP behavior by rewarding those who stay in positions longer — and it is distinct from time-locked liquidity or vesting fee hooks.

Ideas Attempted Only 1–4 Times

These directions have been touched on in past cohorts but remain lightly explored. They represent opportunities to execute more cleanly, integrate new technology (e.g. EigenLayer, Reactive Network), or combine with a second mechanism for a more complete product.

Idea	Times Attempted	Past Example(s)	Opportunity
Delta-neutral hedging via perps	1	xstreamly (UHI4)	The one attempt was AI-heavy and complex. A cleaner, simpler on-chain perp hedge with a clear UX would be competitive.
MEV / LVR redistribution to LPs	3–4	EigenLVR, ROLAID, LVR Auction Hook	Existing attempts focused on MEV capture. None framed it as a yield source for IL offsetting specifically.
Auto-compounding fee reinvestment	2	Auto Compound (UHI1), Prisma (UHI6)	Basic versions exist. Opportunity to add intelligence: only compound when gas is cheap, or route into optimal yield sources.
Fixed-income / tranche LP	1	Fixed/Leverage Yield Hook (UHI3)	One early attempt with limited execution. Strong opportunity for a cleaner, more composable tranche design.
Cross-pool hedging router	0–1	None identified	Listed in the official prompt examples but essentially unbuilt. Strong signal from Uniswap that this direction is desired.
JIT defense for passive LPs	1	ParityTax AMM (UHI5/6)	Addresses JIT concentration indirectly. No hook has built explicit JIT attack detection and routing as an LP protection primitive.
Options collar hook	1–2	Gainswap (UHI3), auction-managed options (UHI1)	Options for LP hedging have been conceptualized but no clean collar mechanism has been executed end-to-end.

Key Takeaways for Facilitators

What has been over-built

The idle liquidity → lending pattern (deploy out-of-range capital to Aave/Morpho) has 5+ implementations across cohorts. Students taking this approach need a strong differentiator: a

specific asset type (LSTs, stables), a novel mechanism on top, or integration with a new protocol.

Where the real white space is

The biggest gaps are in yield tokenization, structured fixed-income products, and proactive IL prevention (gating entry, smoothing realization). These are more TradFi-inspired and require more product thinking than pure mechanism design.

The best UH9 projects will likely combine two things

IL protection alone is a mechanism. Yield generation alone is a mechanism. The projects most likely to win combine both — protecting LPs from downside while generating a new, sustainable yield stream. Think: insurance fund that earns yield on premiums, or a tranche system where the senior tranche earns from lending idle liquidity.